

# Minimizing Crossings

Noah Albright

April 2024

## Introduction

Many of the major problems today in graph theory are how to minimize a graph to some specific criteria or to find some sort of minimum feature in a graph. One of these features for graph visualization is the crossing number of graph  $G$  is defined by

$$Cr(G) = \sum_{i=1}^{m-1} \sum_{j=i+1}^m cross(e_i, e_j)$$

With  $cross(e, e') = \begin{cases} 0 & \text{if there is no common point or the common point is an endpoint of both} \\ 1 & \text{if there is one common point that is an interior point of both} \\ n & \text{if there is overlap between the two edges.} \end{cases}$

The main question of my project is can we predict the rectilinear crossing number of a random graph given the number of edges, nodes, the maximum edge length, minimum edge length, and the closed annulus that contains the graph? We hopefully can use these features to predict the crossing number of a random graph and then use that prediction to try and create an idea to plot a set of edges to a set of points.

## Relevant Works

Now the rectilinear crossing number of a graph has many graph theorists working on conjectures and theorems on the idea. Even the famous mathematician Paul Erdos in the book *Combinatorics, Geometry and Probability: A Tribute to Paul Erdős* which is a collection of Paul Erdos' favorite problems in mathematics that have been solved or is getting worked on at the time. In one chapter it talks about the rectilinear crossing number and also a connection to a question in probabilistic geometry. [4]

Now, as mentioned in the proposal is the paper on the rectilinear crossing number for complete graphs for  $K_5$  through  $K_{27}$ . [1] There is also another paper that there is an upper bound for the rectilinear crossing number for a complete graph  $K_n$  given by

$$Cr(K_n) \leq \frac{7n^4 - 56n^3 + 128n^2 - 48n[\frac{n-7}{3}] + 108}{432}.$$

[3]

Also, a paper similar to what I am doing proposes a deterministic  $n^{2+o(1)}$  time algorithm to predict the rectilinear crossing number of any  $n$ -vertex graph with  $Cr(G)$  having  $(1 + o(1))\overline{Cr}(G)$  with  $\overline{Cr}(G)$  being the minimum rectilinear number of  $G$ . [2] As you can see most of the research on rectilinear crossing numbers has mostly been based heavily on pure math and so one novelty of my system will be using statistics and data science to predict the rectilinear crossing number. Another thing is there is a heavy focus on complete graphs which even though most of my research will be using complete graphs, when I make my model I will focus on any graph with any number of edges that is greater or equal to the number of nodes.

## Methods

My project is broken up into 7 different phases with phase 0 being data collecting and the next five phases looking at each of the features closely to see how they correlate with the crossing number of the graph and if we need to square, log, or some other function on that value to make a linear regression or make sense of the variables.

For phase 0 I first made a c file and shell file to make 1,000 different graphs for each of the other 6 phases. Also throughout each of the other phases, I will normalize the values based on their maximum values so that when we compare them we do not compare a  $10^6$  crossing number to a  $10^{-2}$ . Now why did I do this by the maximum and not the z-score will be given in Appendix D.

For the first phase, I looked at nodes and edges together as there was no way to look at nodes without changing the number of edges at the same time. So we took the two together and made the graphs to look at how they correlated with the crossing number and then made a linear regression using the method of least squares with sklearn's linear model library. I will then verify the regression with scoring against the test data for this phase.

Now for the second phase, I looked at the minimum and maximum edge distance in the graph, and similar to phase 1, I will first look at the scatter plot of each feature to see if they correlate with the crossing number and if they need to do a function. Then once I am finished make and test the linear regression model.

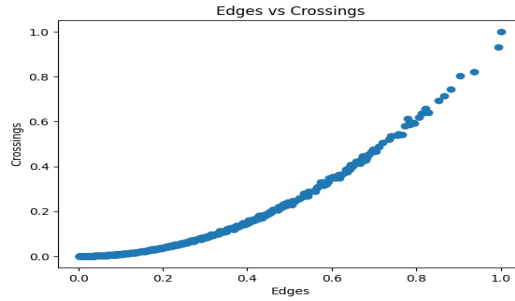
The third and fourth phase is similar to phases 1 and 2 but instead of looking at two features instead, I will look at 1 feature. For phase 3 it was the outer radius and inner radius for phase 4. Now, for phase 5, I will look at the inner and outer radii combined to see if a combination of the two affects the crossing number with the same steps outlined in phases 1 and 2.

For the final phase, I will take the information from the first 5 phases and make my training and test sets on the random graphs. I will then train three

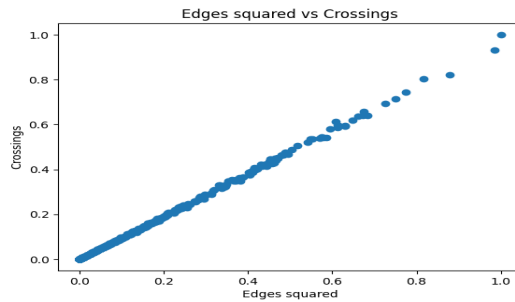
linear regressions using sklearn's LinearRegression, Ridge, and LASSO regressions. For the LinearRegression regression, I will find the best subset on the training data and then test it with that same subset of the test data. For the Ridge and LASSO regressions, I will find the  $\alpha$  values that will make the best regression on our training data and then test them on our testing data. Once all three are finished testing we select the best model based on  $R^2$  and the mean squared error to be our predictor for the crossing number of a random graph.

## Results

Now I will get the results for each phase except phase 0 as it was the data collection phase. For phase 1, I first looked at the comparison between edges and nodes, and as shown in Plot 1, they are very correlated but non-linear. So, as the formula for crossings goes for  $O(\text{edges squared})$ , I squared the edges to get Plot 2 which is a strongly correlated linear scatter plot

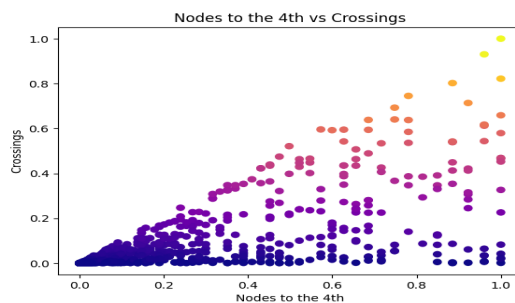


Plot 1: edges vs crossings



Plot 2: edges squared vs crossings

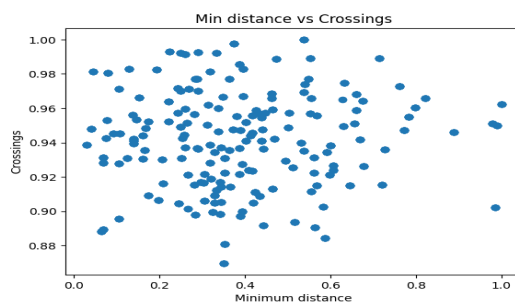
Also for phase 1 for nodes, I got an upper bound correlation which isn't too surprising as stated in the related works is that there is already a known upper bound for a complete graph so using that we will put nodes to fourth and get Plot 3 with edges being the color gradient for the plot.



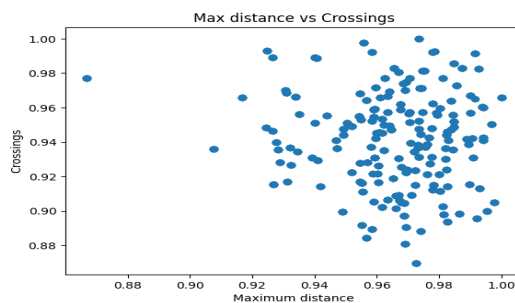
Plot 3: nodes to the fourth vs crossings

Now for phase 1 as you can see from the scatter plots we got a highly correlated linear regression between the edges and nodes with the crossing number with an  $R^2$  of 1.00 on the training data and  $R^2$  of 1.00 on the testing data.

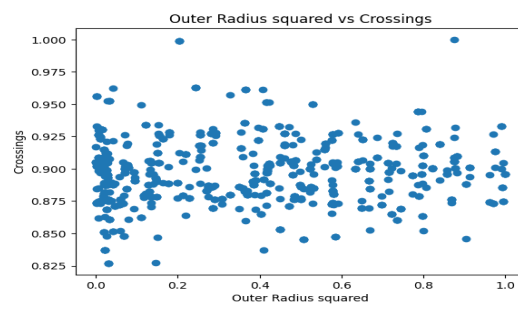
For phases 2 and 3 all my scatter plots given by Plots 4, 5, and 6 have shown no correlation with the data which is also shown in relatively small  $R^2$  scores which are near 0 for the training data and in the negatives for testing data



Plot 4: min distance vs crossings

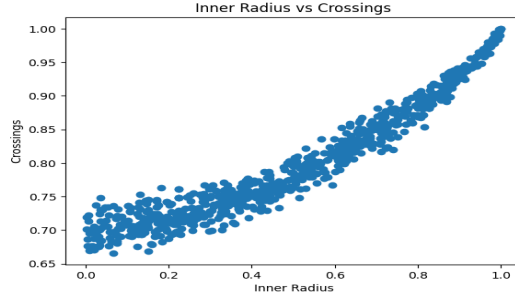


Plot 5: max distance vs crossings

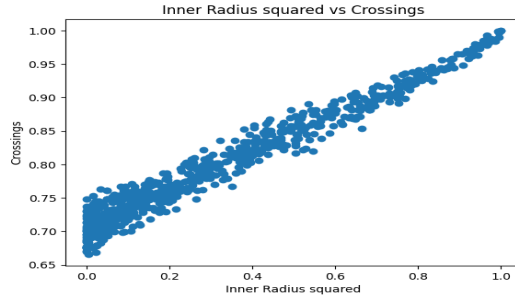


Plot 6: outer radius squared vs crossings

Now the fourth phase on the inner radius is different in this phase the plot for inner radius vs crossing number given by Plot 7 gives a cool-looking horn shape implying some sort of correlation. So as this is the inner radius of the inner circle I squared the inner radius and got a much more linear plot in Plot 8.



Plot 7: inner radius vs crossings

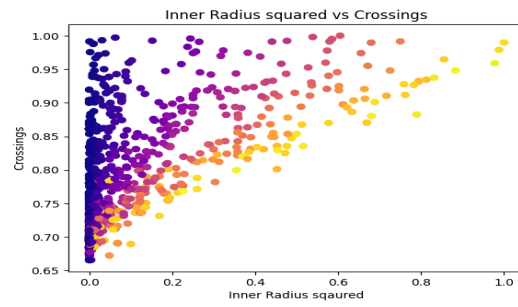


Plot 8: inner radius squared vs crossings

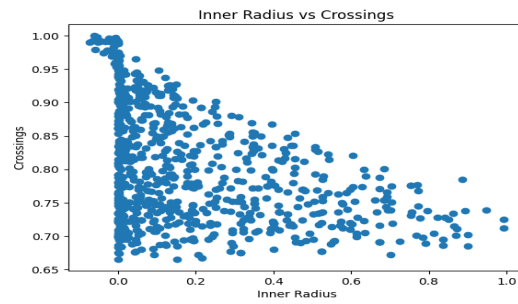
Similar to phase 1, I got a linear regression for the inner radius squared to the crossing number of the graph which got an  $R^2$  of .96 on the training data and an  $R^2$  of .96 on the testing data. I was surprised at that because we thought it would be a little lower looking at the variance when it is lower but we guess it is still very well correlated.

Now for phase 5, we combined phases 3 and 4 which I am noting to show every plot but Plot 9 shows the inner radius squared vs crossings with the color map being the outer radius squared. The other plot that we are going to show is the outer radius squared minus the inner radius squared vs crossings in Plot 10 as we going to add the outer radius minus the inner radius to the features for phase 6.

For phase 5, as we can see from the plots there is not a strong correlation but still some correlation which is why we got an  $R^2$  of .54 on the training data and  $R^2$  of .56 on the testing data meaning they are collaterated but not as much



Plot 9: inner radius squared vs crossings



Plot 10: outer radius squared - inner radius squared vs crossings

as phase 1 and phase 4.

Now for the finishing phase, we trained three different linear regression models from sklearn LinearRegression, Ridge, and LASSO. For LinearRegression we looked at every non-empty subset of features to find the subset of features that gave the greatest  $R^2$  value which was the subset of nodes, edges, inner radius, and maximum distance with  $R^2 = .99$ . For Ridge, we tried to find the best  $\alpha$  value that gave the maximum  $R^2$  which was  $\alpha = 0.1$  with  $R^2 = .99$ . Now for LASSO, we try to find the best  $\alpha$  value that gave the minimum mean squared error which was  $\alpha = 58$  with a mean squared error of .00.

For the final part, I tested each of the three trained linear models on the testing data for phase 6. For LinearRegression, I got an  $R^2$  of .99 and a mean squared error of .00; For Ridge, I got an  $R^2$  of .99 and a mean squared error of .00; And for LASSO, I got an  $R^2$  of  $-0.011$  and a mean squared error of 0.03. So the overall best regression is technically LinearRegression but it is very close to Ridge so I will state both of their functions. Let  $n$  = nodes,  $e$  = edges,  $r$  = inner radius,  $R$  = outer radius,  $m$  = minimum distance, and  $M$  = maximum distance; then

$$\text{LinearRegression } Cr(G) = -0.003n^4 + 1.031e^2 + 0.031r^2 - 0.016M$$

and

$$\text{Ridge } Cr(G) = 0.007n^4 + 1.006e^2 + 0.014r^2 - 0.001R^2 + 0.000m + 0.001M - 0.015(R^2 - r^2).$$

## Conclusion

Going through this project was a lot of work, especially for many features that were not very correlated with the rectilinear crossing number of a graph. The one thing I will conclude from this project is that for any graph it is probably best to look at the number of edges and squaring it to predict the rectilinear crossing number of a graph. It also explains why there is not much data science on this topic either as there is nothing much that data can give you besides look at how many edges you have.

Now one thing I do want to look at further from my data is the maximum distance and inner radius features and how they compare with the rectilinear crossing number. Regarding maximum distance, I want to look at it further as in phase 2, it looks like it is not correlated at all to the rectilinear number but at the end, it is in the best subset of features so maybe if you look at different distances or a wider range of distances we can see some correlation for it. Now for the inner radius, it was correlated but I stated in the paper and the notebook that as it increases the less variances it is so maybe we can research further why it is. Also just as I finish up this paper another idea is to divide it with the outer radius to see if that is in any way correlated to the rectilinear crossing number.



Overall, this project was a lot of fun but stressful as basically, the answer was clear based on the research done on this subject but I had some ideas I wanted to try and find an answer on those ideas. So, in the fall semester, I am taking a Graph theory course, and if we ever have a topic on the crossing numbers of a graph and maybe apply more knowledge later.

## Appendices

### Appendix A - makeData.c and makeData.sh

You will notice that I have included a makeData.c and makeData.sh files with my notebook which as the name of the files suggests are the files that made all my data for this project. Now you don't have to run but if you want to know how I got my data then they are included with my notebook submission (also I don't want to 500 lines of c code in my notebook). Also, some of the appendices are about some stuff that happens in makeData.c.

### Appendix B - makeData.c random edges

So in the makeData.c you will see the comment in my makeEdges function to explain why the deterministic way that I made the edges is still random as it is not obvious why it will be (or at least to me anyway). The edges are random as the nodes (points) that make up the edges are random and thus even in the deterministic way that I made them it will be random.

For example, take me throwing  $n$  different dice with each having a random amount of sides then we remove the last half of dice that we threw. Now we selected  $n/2$  dice which is random as we didn't know which die would be going at the  $k$ th position for  $1 \leq k \leq n$  so as this example shows we can have the process to remove be deterministic while it still creates a random selection.

### Appendix C - makeData.c getCross

Also, in the makeData.c you will see in the function comment on the crossings, cross, and getCross function that more on those functions will be given in the paper. Now, the first should be self-explanatory by the introduction to the paper so I will not be going over those two, instead, we will focus on getCross and how it finds which of the three options our two edges are.

The main idea is that a rectilinear cross is just solving a system of linear equations between the two edges. For the first part of my code, it just checks if the edges are first parallel and if they lie on the same line. Between those two and some distance checking, we can see if they have some sort of overlap option 2 or none at all 0. After passing through the first part we can guarantee a single unique solution to the system. Now there are many ways to solve the linear equation like row reduction or Cramer's rule (tried this originally) but row reduction is hard to do for a computer and Cramer's Rule was not working with my code. So I just solved the system just as I would have in high school

which let  $a_1x + b_1x = c_1$  and  $a_2x + b_2x = c_2$  be the two linear equations with assume that  $a_1 \neq 0$ . Then the unique solution to the system is

$$y = \frac{c_2 - c_1 * \frac{a_2}{a_1}}{b_2 - b_1} * \frac{a_2}{a_1}$$

$$x = \frac{c_1 - b_1y}{a_1}.$$

Now, for  $a_1 = 0$ , we just flip the order we solve for x and y which will result in the code given in `getCross`.

## Appendix D - Why z-score fails

Earlier in my paper I mentioned that I normalized my data by dividing by the maximum instead of using the z-score and I will explain that here. If you look through my code you will see that whenever I change a feature I will mostly square it because that is what the data provides I do. So, if I first wanted to normalize my data and then look at the graphs I would need a normalization metric that when I multiply it is the same as multiplying that data point, and for a z-score that is not the case. The reason why is that it subtracts the mean so  $z^2$  is the value minus the mean square which is not equal to the squared value minus the squared mean and thus  $z^2$  is not the valued squared.

## References

- [1] Bernardo M. Ábrego et al. “The maximum number of halving lines and the rectilinear crossing number of  $K_n$  for  $n \leq 27$ ”. In: *Electronic Notes in Discrete Mathematics* 30 (2008). The IV Latin-American Algorithms, Graphs, and Optimization Symposium, pp. 261–266. ISSN: 1571-0653. DOI: <https://doi.org/10.1016/j.endm.2008.01.045>. URL: <https://www.sciencedirect.com/science/article/pii/S1571065308000462>.
- [2] Jacob Fox, János Pach, and Andrew Suk. “Approximating the rectilinear crossing number”. In: *Computational Geometry* 81 (2019), pp. 45–53. ISSN: 0925-7721. DOI: <https://doi.org/10.1016/j.comgeo.2019.04.003>. URL: <https://www.sciencedirect.com/science/article/pii/S0925772119300537>.
- [3] H.F Jensen. “An upper bound for the rectilinear crossing number of the complete graph”. In: *Journal of Combinatorial Theory, Series B* 10.3 (1971), pp. 212–216. ISSN: 0095-8956. DOI: [https://doi.org/10.1016/0095-8956\(71\)90045-1](https://doi.org/10.1016/0095-8956(71)90045-1). URL: <https://www.sciencedirect.com/science/article/pii/0095895671900451>.
- [4] H.S. Wilf. “On Crossing Numbers, and some Unsolved Problems”. In: *Combinatorics, Geometry and Probability: A Tribute to Paul Erdős*. Ed. by Béla Bollobás and Andrew Editors Thomason. Cambridge University Press, 1997, pp. 557–562.